

HONG KONG BAPTIST UNIVERSITY

Audio and Speech Processing

Practical Assignment 2

Introduction

This practical assignment is designed to assess students' understanding and programming skills in conducting audio source separation tasks. Students will be tested on their replication and implementation of building traditional and deep learning methods, including *ICA*, *RPCA*, *Conv-TasNet*, *DPRNN*, and *DEMUCS*. Bonus scores will be provided to students for their innovative programming on an extra task with additional feature construction. They are also required to report results by showing critical thinking in interpreting the results.

Data

Like the sample data demonstrated in the lab, two sets of data are provided. Each student will work on the data according to the last digit in your student ID. Please refer to *Table 1 in Appendix A* to find the data to work on

Program Code Template

The basic Python program code is provided for your reference in the tutorial notebook "Lab2 – Sound Separation.ipynb". You should complete the tasks in the document "Lab2 – Exercises.ipynb". All the tasks can be implemented based on the given code in the tutorial notebook with slight adjustments, such as replicating certain functions, changing parameters, or features. The evaluation metrics are SDR, SIR, and SAR

Programming Environment

All provided programs were tested in *Python=3.10.20* and *PyTorch=2.13.0*. To avoid any package incompatibility, you are encouraged to complete the tutorial and tasks in the same package version. Google Colab and PyCharm can be the platforms for you to execute the program

Tasks

- **Task 1:** ICA implementation. Combine the provided data to create a 4-channel audio mixture. Use the *FastICA* algorithm to separate this mixture into four distinct sources. Calculate the evaluation metrics for the separated sources. For each separated source, generate visualizations of both its signal waveform and its spectrogram
- **Task 2:** RPCA with different parameters. (1) Apply the *RPCA* to the given data with different combinations of the parameters *nFFT* and *hopLength*, then compute the evaluation metrics for each combination. Discuss how different combinations of these two parameters affect the performance. (2) With fixed *nFFT* and *hopLength*, apply *RPCA* by trying different values of *lmbda* to achieve the required baseline for the evaluation metrics. Discuss the impact of varying *lmbda* on the performance
- **Task 3:** Conv-TasNet. (1) Use the pre-trained *Conv-TasNet* model to separate the given audio. Display the separated sources. Plot the signals and spectrograms. (2) Complete the full process of building and training a *Conv-TasNet* model in a given notebook "Lab2 – TasNet.ipynb"

- **Task 4:** DPRNN to enhance speech. Use the provided script to record and save your own speech locally. Mix your recorded speech with the given noise file to create a noisy audio sample. Load the pre-trained DPRNN model *JorisCos/DPRNNTasNet-ks2_Libri1Mix_enhsingle_16k* to perform speech enhancement on the noisy audio. Play back the enhanced result and plot spectrograms of both the noisy and enhanced audio for comparison
- ***Extra/bonus Task 1:** TBC.

Submission

Submit ONE *.ipynb* file containing all your program codes for tasks. Rename the *ipynb* file to your name, e.g. ChanTaiMan.*ipynb*. Submit the bonus problem files separately.

Appendix A

You should work on one of the corresponding subset data as listed in Table 1.

Last digit of student ID	Name of subset data
0-4	xxx_A.zip / xxx_A.wav / xxx_A.mp3
5-9	xxx_B.zip / xxx_B.wav / xxx_B.mp3

Table 1 - Mapping of Student ID and Subset Number